

UNIVERSITY OF MAURITIUS

FACULTY OF INFORMATION, COMMUNICATION AND DIGITAL TECHNOLOGIES



SECOND SEMESTER EXAMINATIONS

JULY/AUGUST 2020

PROGRAMME	BSc (Hons) Information Systems		
MODULE NAME	Information Systems Security		
DATE	Thursday 30 July 2020	MODULE CODE	CSE3118Y
TIME	10.30-12.30 hrs	DURATION	2 Hours
NO. OF QUESTIONS SET	5	NO. OF QUESTIONS TO BE ATTEMPTED	4

INSTRUCTIONS TO CANDIDATES

Answer any 4 questions.

All questions carry equal marks.

Question 1

Answer any 4 questions.

All questions carry equal marks.

- (a) Name five main concepts aim at in a secure system. **[5 marks]**
- (b) Using suitable examples of your own explain the concepts mentioned in (a) and their respective protection mechanism. **[3 marks]**
- (c) Briefly explain the terms following terms:
- (i) Hacking
 - (ii) Virus-
 - (iii) Malware
 - (iv) Scareware
 - (v) Cracking
- [5 marks]**
- (d) All staffs in an organization should be involved in the building of a secure working environment. How should management participate in that endeavor? **[3 marks]**
- (e) Comment on the need for a security policy **[4 marks]**
- (f) As a security expert what are the steps you would take to ensure an adequate security level in an organization. **[5 marks]**
- (Total: 25 Marks)**

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Question 2

- (a) Distinguish between **unconditional** and **computational** security characteristic of a cipher.
[4 marks]
- (b) Differentiate between symmetric and public key cryptosystems.
[6 marks]
- (c) Comment on the importance and operation of the Feistel network.
[4 marks]
- (d) Explain using a labeled diagrammed the operation of a full fledged hybrid cipher system which also caters for integrity.
[6 marks]
- (e) From the section above (d) we can also add non repudiation through the use of signature. Explain the purpose of using digital signature and on what basis does it work?

[5 marks]
(Total: 25 Marks)

Question 3

- (a) Explain the term firewall and use suitable examples of your own to supplement your answer.
[2 marks]
- (b) Name three network devices that can act as a firewall
[3 marks]
- (c) Differentiate between the firewall policies that exist.
[4 marks]
- (d) Comment on the following statement “though we have implemented the most expensive firewall we are still having security problems.”
[6 marks]
- (e) Explain operation of the widely used DMZ architecture.
[7 marks]
- (f) Briefly describe the different sources through which an intruder may get access to a system.

[3 marks]
(Total: 25 Marks)

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Question 4

- (a) Differentiate between steganography and watermarking?
[4 marks]
- (b) Name three uses of watermarking and three uses of steganography.
[6 marks]
- (c) Differentiate between spatial and frequency domain watermark.
[4 marks]
- (d) A watermark system consists of three main phases. Using a labeled diagram explain operation of watermark system.
[6 marks]
- (e) Computer forensics relies on the three basic methodologies when gathering evidences. Name these three methodologies and show their sequence of occurrence.
[3+2 marks]

Question 5

- (a) Explain the different vertex of the authentication triangle. Use suitable examples of your own to demonstrate its use.
[6 marks]
- (b) Differentiate between soft and hard biometric in terms of ease of use, performance and number of application in use.
[4 marks]
- (c) List five important properties which an element should possess before its use in biometric authentication.
[5 marks]
- (d) Explain the operation of a biometric system in verification mode. Use block diagrams to simplify the steps.
[5 marks]
- (e) Comment on three IT laws prevailing in Mauritius. Briefly comment on the need for these laws.
[3*1 + 2 marks]

END OF QUESTION PAPER